



Image Processing Lab 6



Convolution Operation

Source layer

5	2	6	8	2	0	1	2
4	3	4	5	1	9	6	3
3	9	2	4	7	7	6	9
1	3	4	6	8	2	2	1
8	4	6	2	3	1	8	8
5	8	9	0	1	0	2	3
9	2	6	6	3	6	2	1
9	8	8	2	6	3	4	5

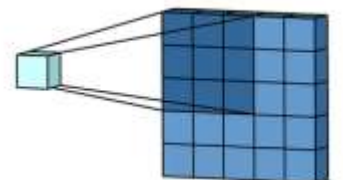
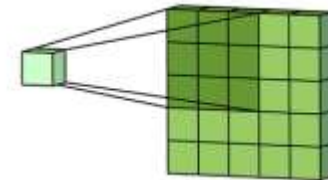
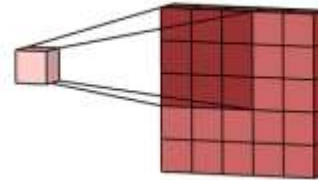
Convolutional
kernel

-1	0	1
2	1	2
1	-2	0

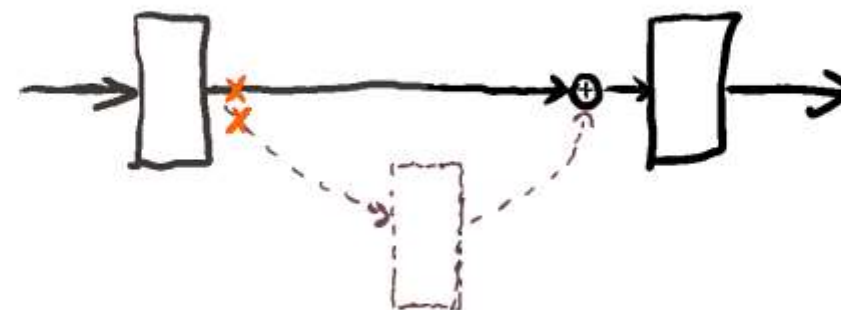
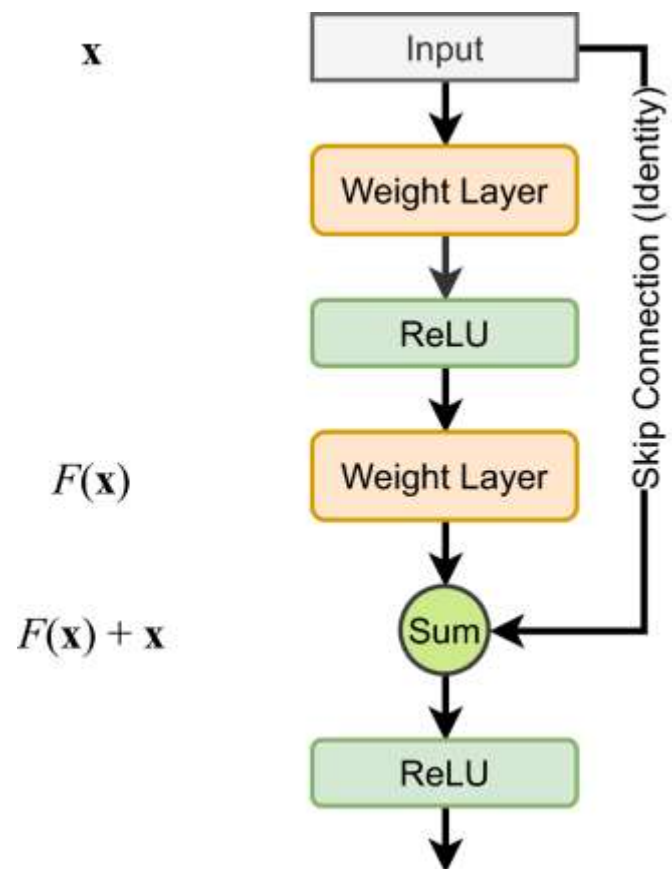
Destination layer

	5						

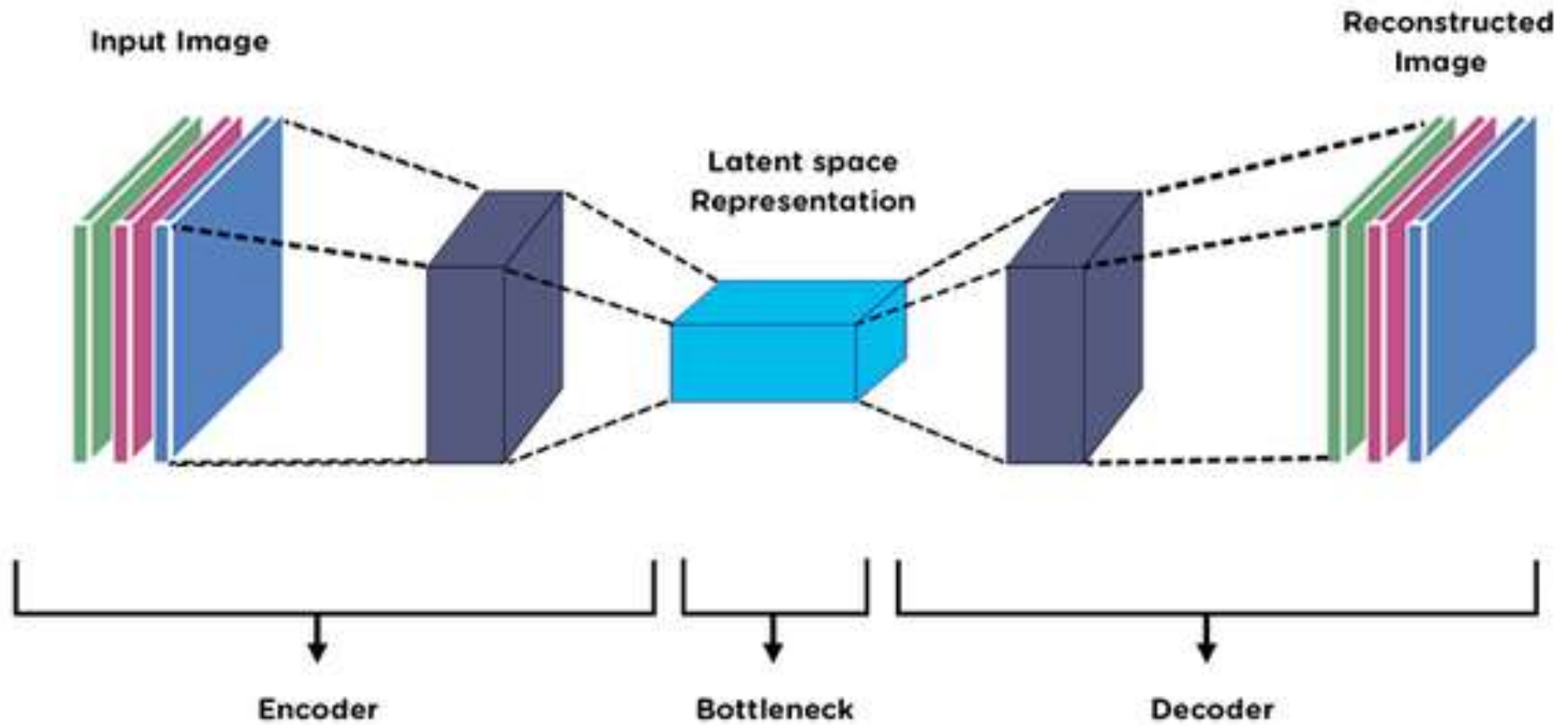
$$\begin{aligned} &(-1 \times 5) + (0 \times 2) + (1 \times 6) + \\ &(2 \times 4) + (1 \times 3) + (2 \times 4) + \\ &(1 \times 3) + (-2 \times 9) + (0 \times 2) = 5 \end{aligned}$$



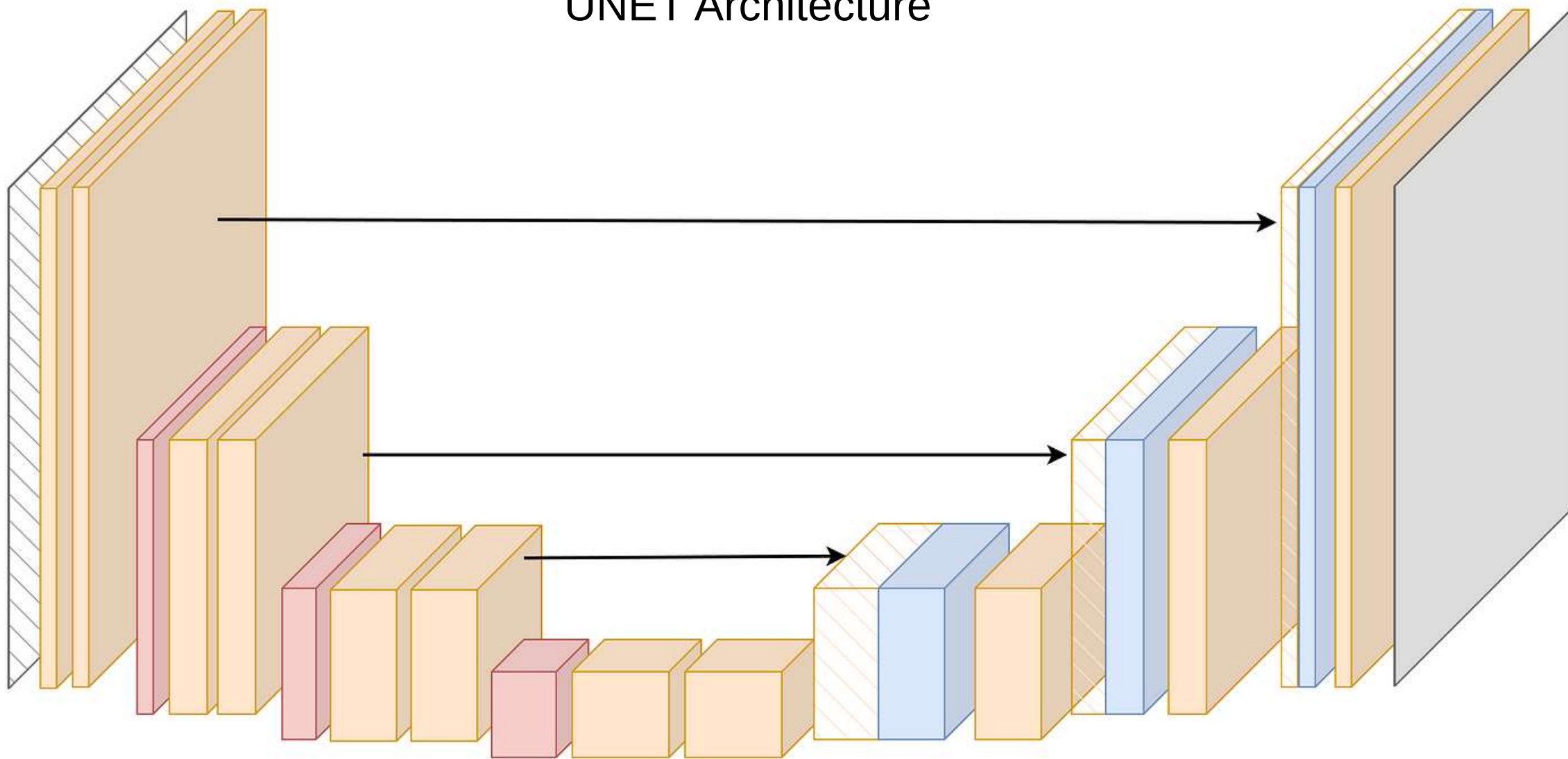
Skip Connections

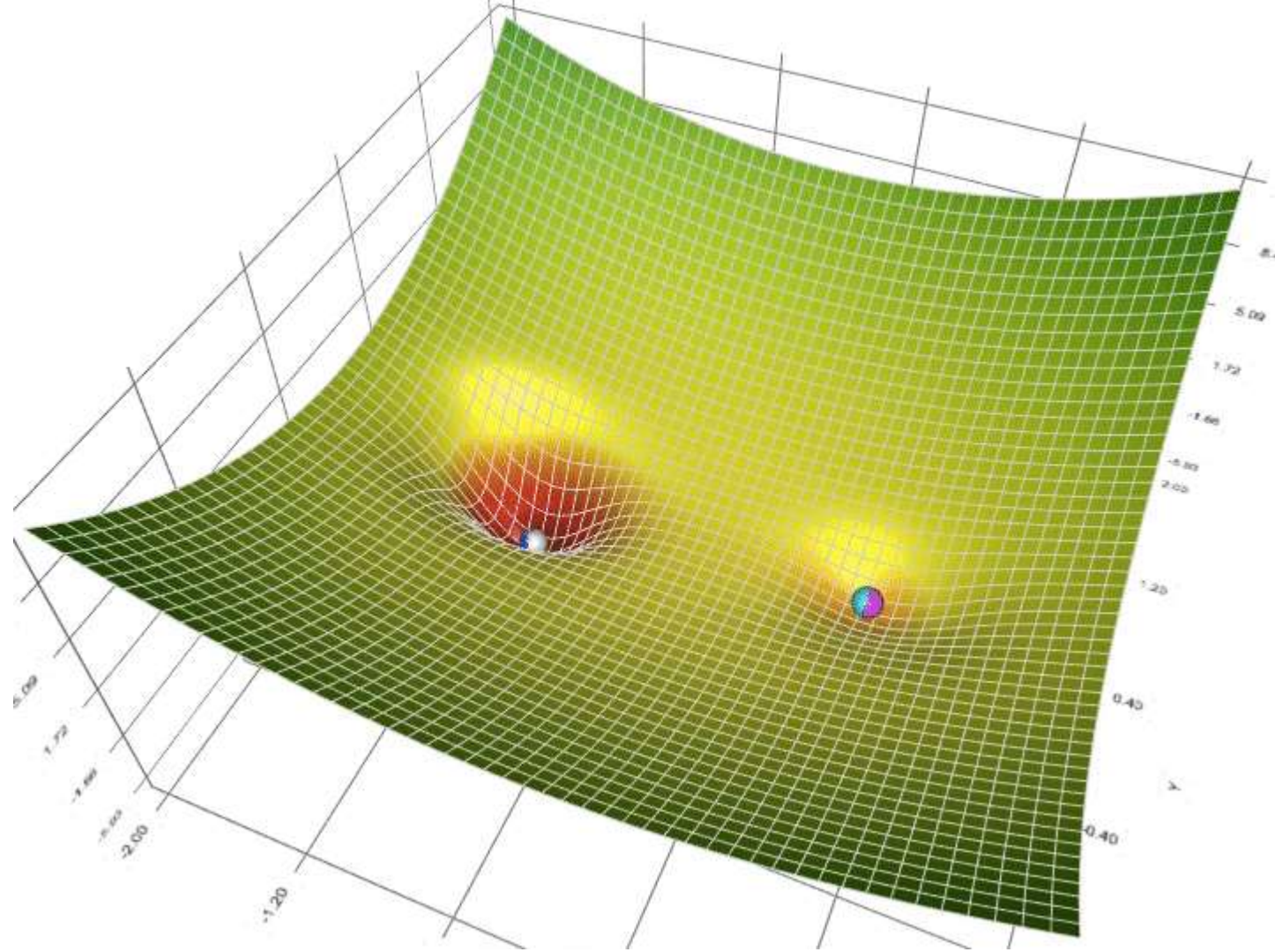


AutoEncoder



UNET Architecture





```
optimizer=tf.keras.optimizers.Adam()
```

Gradient Descent

Our Goal

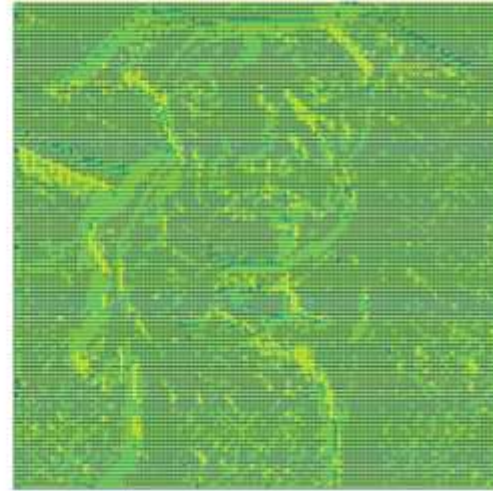
Input_Image



True_Mask



Predicted_Mask



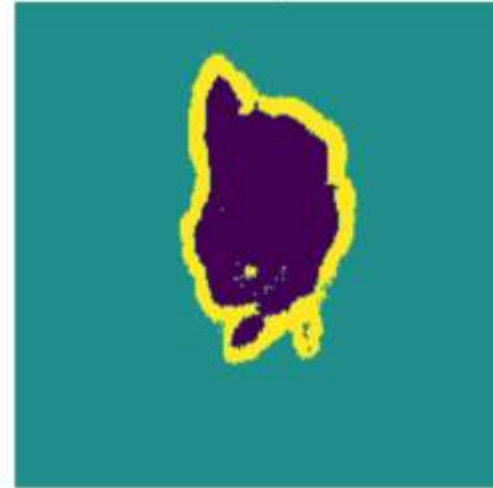
Input_Image



True_Mask



Predicted_Mask

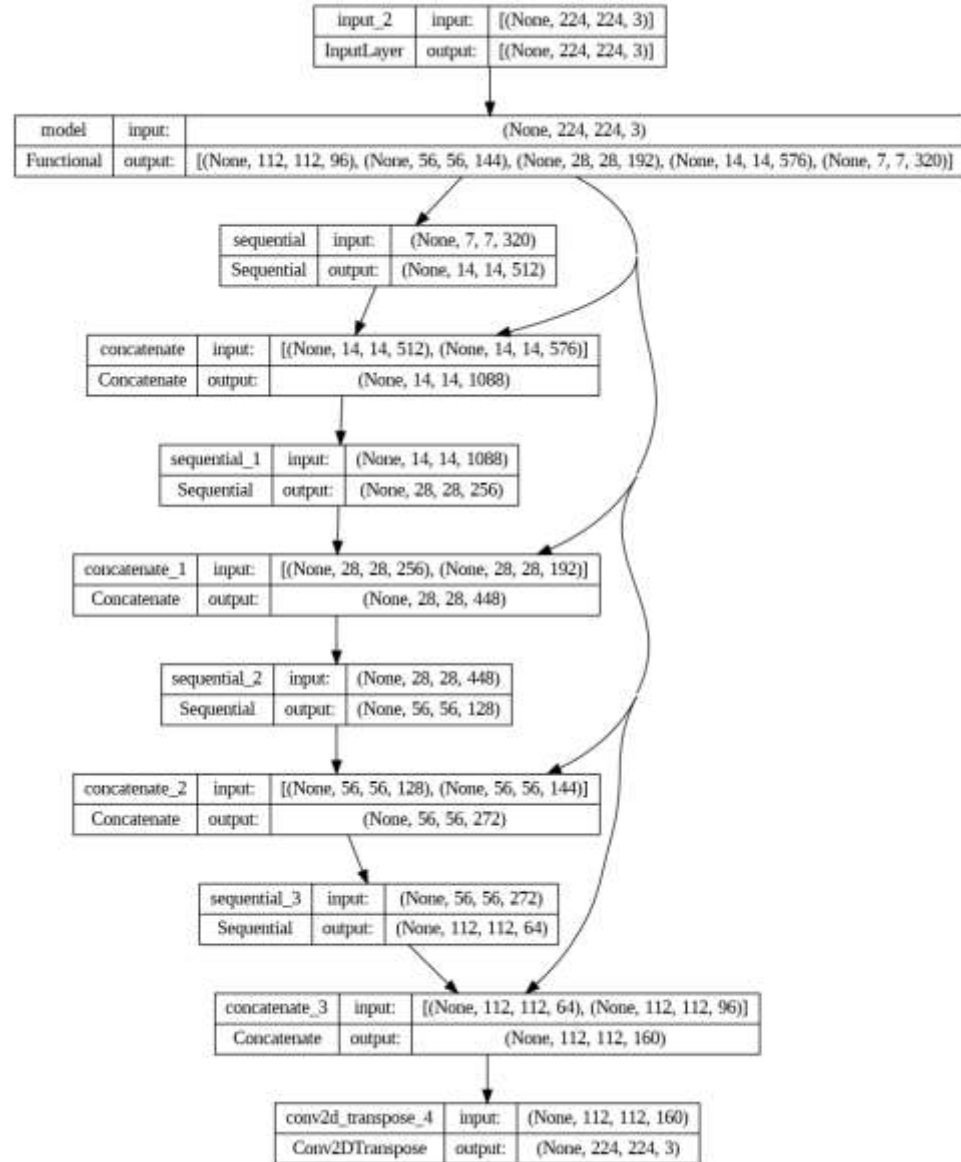


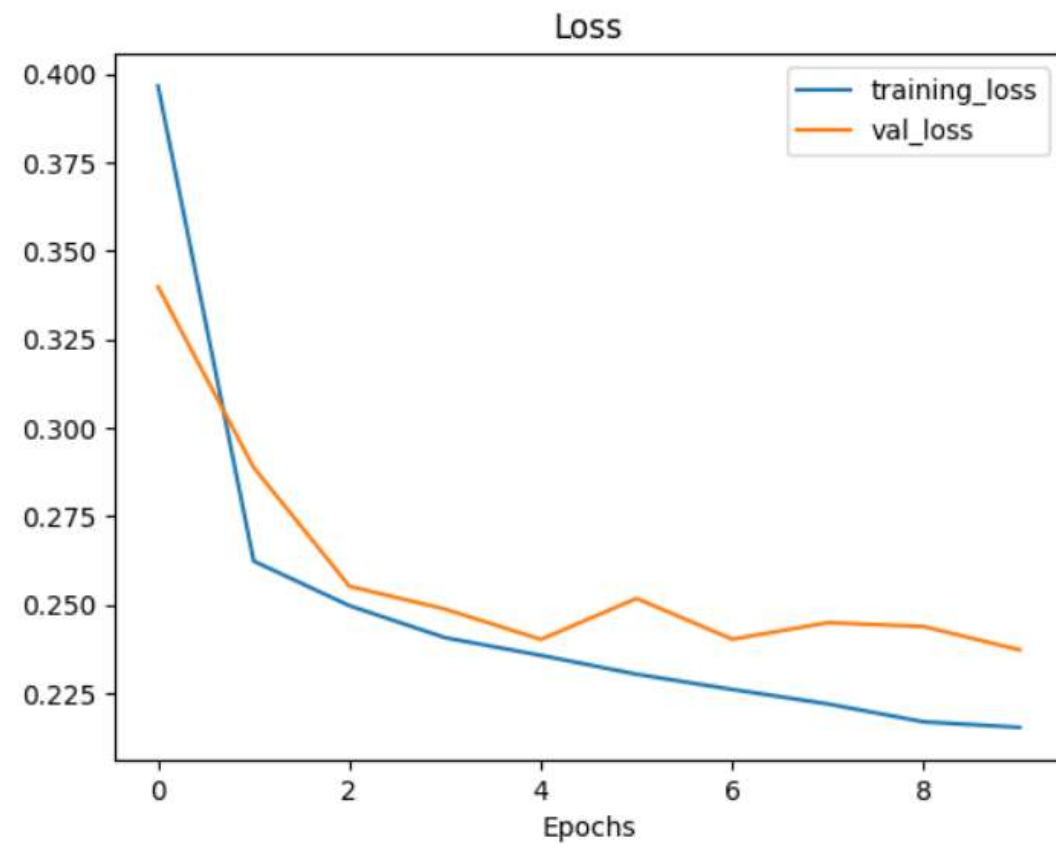
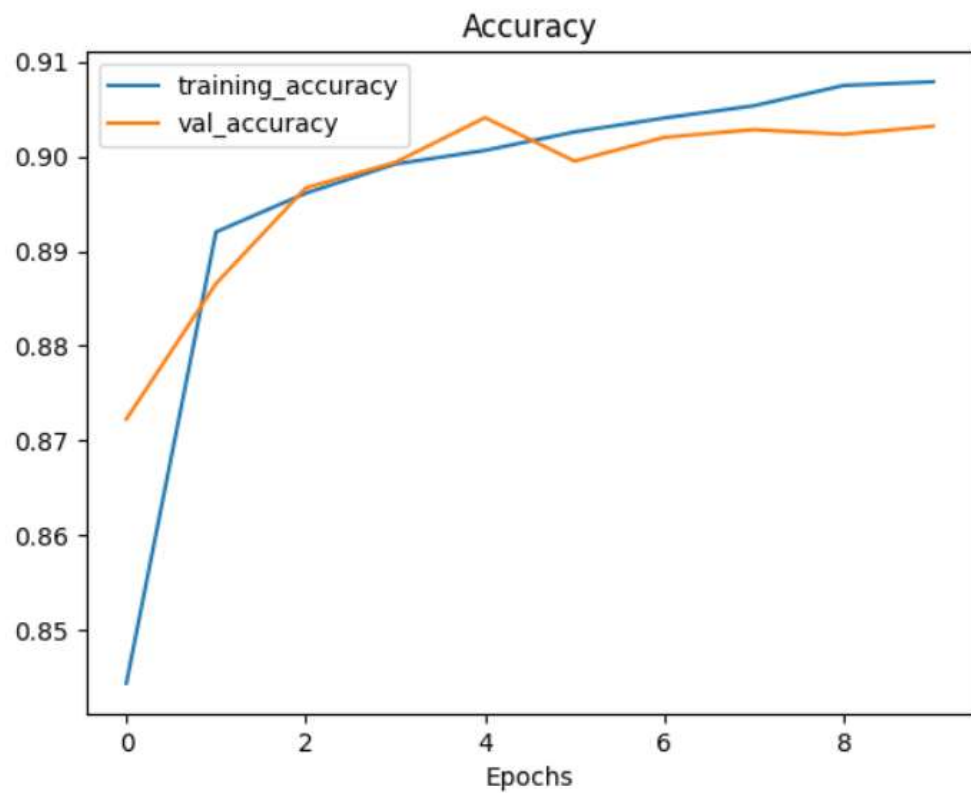


#information ds_info

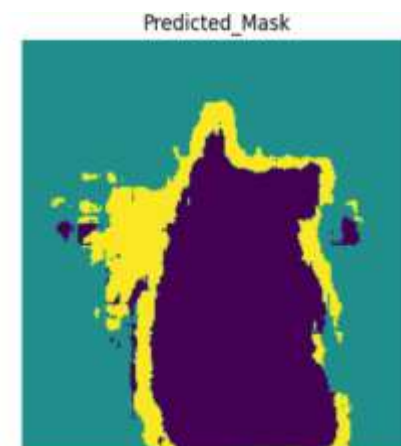
```
#download the dataset
(train_data, test_data), ds_info = tfds.load('oxford_iiit_pet:3.*.*',
with_info=True,
split=['train', 'test'])
```

```
tfds.core.DatasetInfo(
  name='oxford_iiit_pet',
  full_name='oxford_iiit_pet/3.2.0',
  description="""
The Oxford-IIIT pet dataset is a 37 category pet image dataset with roughly 200
images for each class. The images have large variations in scale, pose and
lighting. All images have an associated ground truth annotation of breed.
""",
  homepage='http://www.robots.ox.ac.uk/~vgg/data/pets/',
  data_dir=PosixGPath('/tmp/tmpk1xdso35tfds'),
  file_format=tfrecord,
  download_size=773.52 MiB,
  dataset_size=774.69 MiB,
  features=FeaturesDict({
    'file_name': Text(shape=(), dtype=string),
    'image': Image(shape=(None, None, 3), dtype=uint8),
    'label': ClassLabel(shape=(), dtype=int64, num_classes=37),
    'segmentation_mask': Image(shape=(None, None, 1), dtype=uint8),
    'species': ClassLabel(shape=(), dtype=int64, num_classes=2),
  })),
  supervised_keys=('image', 'label'),
  disable_shuffling=False,
  splits={
    'test': <SplitInfo num_examples=3669, num_shards=4>,
    'train': <SplitInfo num_examples=3680, num_shards=4>,
  },
  citation="""@InProceedings{parkhi12a,
  author      = "Parkhi, O. M. and Vedaldi, A. and Zisserman, A. and Jawahar, C.~V.",
  title       = "Cats and Dogs",
  booktitle   = "IEEE Conference on Computer Vision and Pattern Recognition",
  year        = "2012",
}""",
)
```



Prediction on Test set





Testing on Random Image from the Internet

List of Datasets available on TFDS

```
[ ] tfds.list_builders()
```

<https://www.tensorflow.org/datasets/catalog/overview>

```
['abstract_reasoning',  
 'accentdb',  
 'aeslc',  
 'aflw2k3d',  
 'ag_news_subset',  
 'ai2_arc',  
 'ai2_arc_with_ir',  
 'aloha_mobile',  
 'amazon_us_reviews',  
 'anli',  
 'answer_equivalence',  
 'arc',  
 'asqa',  
 'asset',  
 'assin2',  
 'asu_table_top_converted_externally_to_rlds',  
 'austin_buds_dataset_converted_externally_to_rlds',  
 'austin_sailor_dataset_converted_externally_to_rlds',  
 'austin_sirius_dataset_converted_externally_to_rlds',  
 'bair_robot_pushing_small',  
 'bc_z',  
 'bccd',  
 'beans',
```


Objective:

*Given all these examples to you, your job in this lab is to **improve the performance** of the model compared to the baseline model.*

Options to explore:

- Train for more epochs
- Modify learning rate (LR)
- Modify the Optimizer (SGD, Adam, ...)
- Learning Rate Schedulers (Cosine Annealing)

Also, You're encouraged to change the dataset from oxform iit pet to any other dataset online available on the internet or on tfds. You can further compare the loss and accuracy of this baseline model vs your improved models based on the changes you have made.

You can further analyse other performance matrices other than accuracy such as, precision, recall, F1 score.

You need to submit a comprehensive report along with results comparing this baseline model and your models highlighting your methodology. If model 1 (based on some changes) does not improve the model performance you can still mention that in your report but justify your answer why it did not work. You can check the list of datasets available on tfds using this code,